

#NumPy Tasks

Task 1

```
print("Task 1",)
import numpy as np
arr = np.arange(1, 11)
print(arr)
```

Task 2

```
print("\nTask 2")
import numpy as np
arr = np.array([10, 20, 30, 40, 50])
print("Mean:", np.mean(arr))
print("Median:", np.median(arr))
print("Standard Deviation:", np.std(arr))
```

Task 3

```
print("\nTask 3")
import numpy as np
random_numbers = np.random.rand(10)
print(random_numbers)
```

Task 4

```
print("\nTask 4")
import numpy as np
arr = np.arange(1, 9)
reshaped = arr.reshape(2, 4)
print(reshaped)
```

Task 5

```
print("\nTask 5")
import numpy as np
arr = np.array([5, 12, 7, 20, 3])
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))
```

Task 6

```
print("\nTask 6")
import numpy as np
zeros = np.zeros(5)
ones = np.ones(5)
print("Zeros:", zeros)
print("Ones:", ones)
```

===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 33 TASK.

Task 1

[1 2 3 4 5 6 7 8 9 10]

Task 2

Mean: 30.0

Median: 30.0

Standard Deviation: 14.142135623730951

Task 3

[0.26694188 0.99712049 0.22238356 0.35991179 0.11729505 0.96330734
0.60586571 0.11822241 0.60867594 0.39946502]

Task 4

[[1 2 3 4]
[5 6 7 8]]

Task 5

Maximum: 20

Minimum: 3

Task 6

Zeros: [0. 0. 0. 0. 0.]

Ones: [1. 1. 1. 1. 1.]

```

#Pandas Tasks

# Task 1
print("Task 1")
import pandas as pd
data = {
    "Name": ["Aman", "Ravi", "Sita", "John"],
    "Marks": [85, 90, 78, 88]
}
df = pd.DataFrame(data)
print(df)

# Task 2
print("\nTask 2")
import pandas as pd
data = {"Name": ["A", "B", "C"], "Marks": [70, 80, 90]}
df = pd.DataFrame(data)
df["Grade"] = ["B", "A", "A"]
print(df)

# Task 3
print("\nTask 3")
import pandas as pd
data = {"Name": ["A", "B", "C"], "Marks": [70, 90, 80]}
df = pd.DataFrame(data)
sorted_df = df.sort_values(by="Marks")
print(sorted_df)

# Task 4
print("\nTask 4")
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head(25))

# Task 5
print("\nTask 5")
import pandas as pd
data = {"A": [1, 2, None, 4], "B": [5, None, 7, 8]}
df = pd.DataFrame(data)
df = df.dropna()
print(df)

```

```
# Task 6
print("\nTask 6")
import pandas as pd
data = {"Marks": [50, 60, None, 80, 90]}
df = pd.DataFrame(data)
df["Marks"] = df["Marks"].fillna(df["Marks"].median())
print(df)
```

```

>>
===== RESTART: C:/Users/krsb/OneDrive/
Task 1
      Name  Marks
0  Aman      85
1  Ravi      90
2  Sita      78
3  John      88

Task 2
      Name  Marks  Grade
0    A      70      B
1    B      80      A
2    C      90      A

Task 3
      Name  Marks
0    A      70
2    C      80
1    B      90

Task 4
      Name  Marks
0  Aman      85
1  Ravi      90
2  Sita      78
3  John      88
4  Rohit      76
5  Priya      91

Task 5
      A      B
0  1.0  5.0
3  4.0  8.0

Task 6
      Marks
0  50.0
1  60.0
2  70.0
3  80.0
4  90.0
>>

```

#Matplotlib Tasks

Task 1

```
print("Task 1")
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr"]
sales = [200, 300, 250, 400]
plt.plot(months, sales)
plt.show()
```

Task 2

```
print("\nTask 2")
import matplotlib.pyplot as plt
students = ["A", "B", "C", "D"]
marks = [80, 90, 70, 85]
plt.bar(students, marks)
plt.show()
```

Task 3

```
print("\nTask 3")
import matplotlib.pyplot as plt
height = [150, 160, 170, 180]
weight = [50, 60, 70, 80]
plt.scatter(height, weight)
plt.show()
```

Task 4

```
print("\nTask 4")
import matplotlib.pyplot as plt
ages = [20, 22, 21, 23, 24, 22, 25, 26]
plt.hist(ages)
plt.show()
```

Task 5

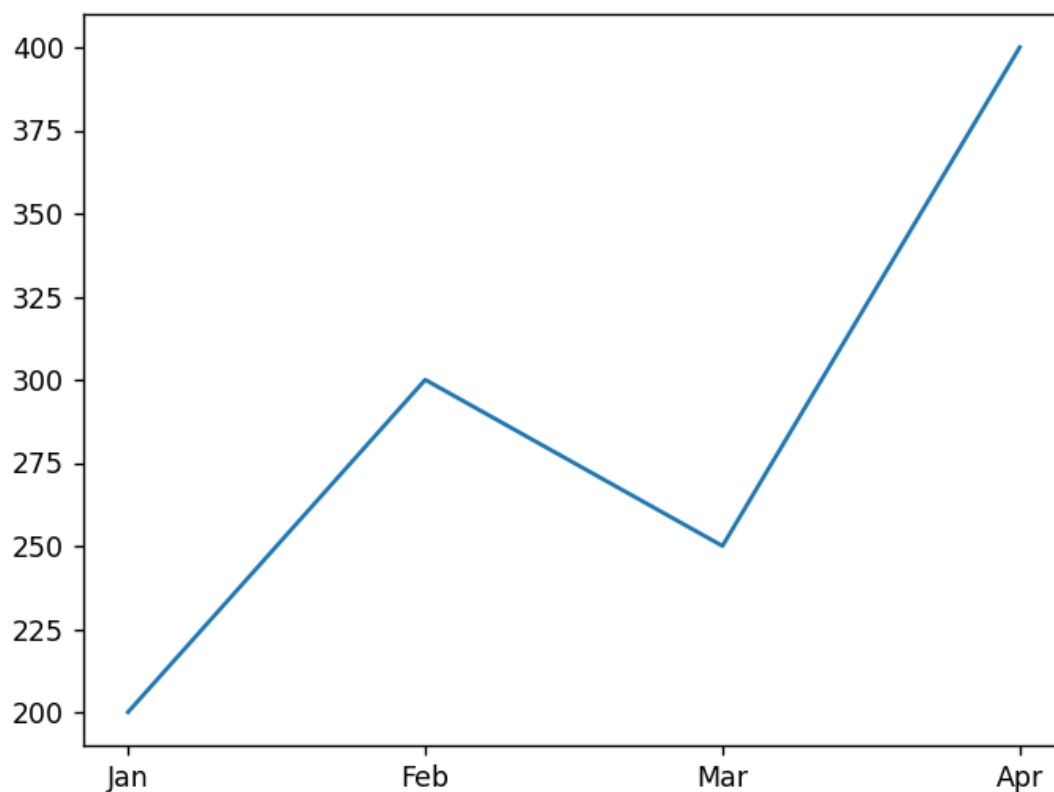
```
print("\nTask 5")
import matplotlib.pyplot as plt
x = [1, 2, 3, 4]
y = [10, 20, 25, 30]
plt.plot(x, y, label="Sales")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.legend()
plt.show()
```

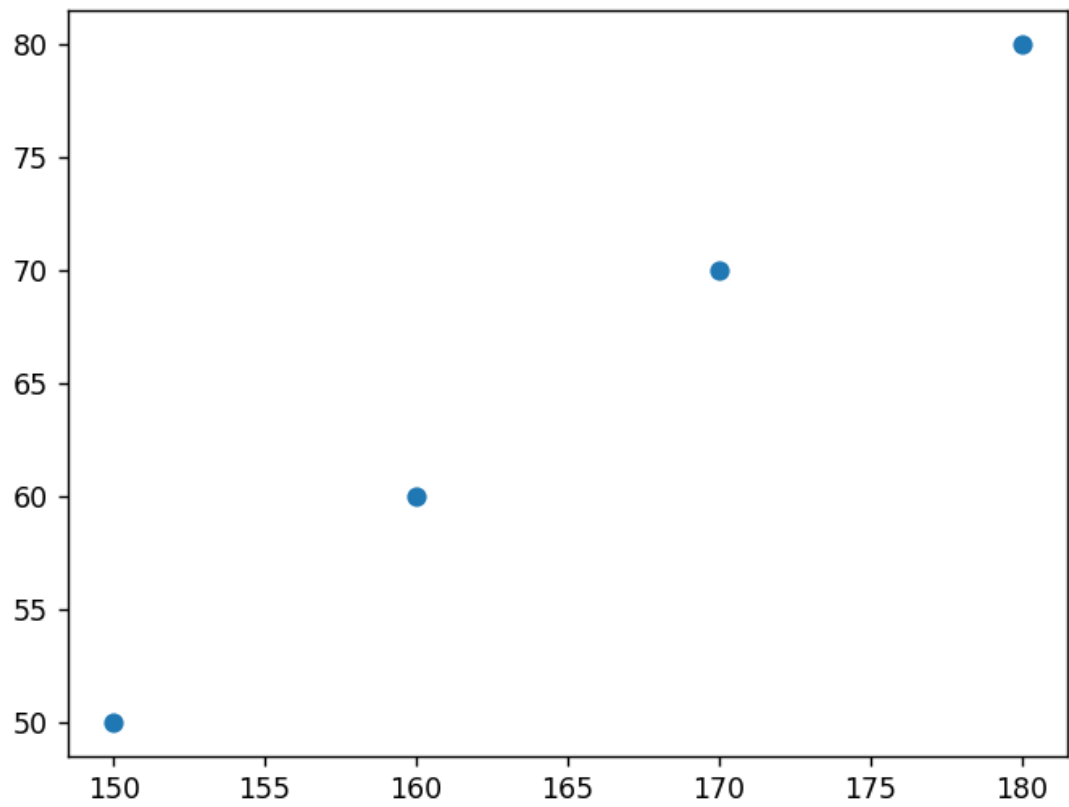
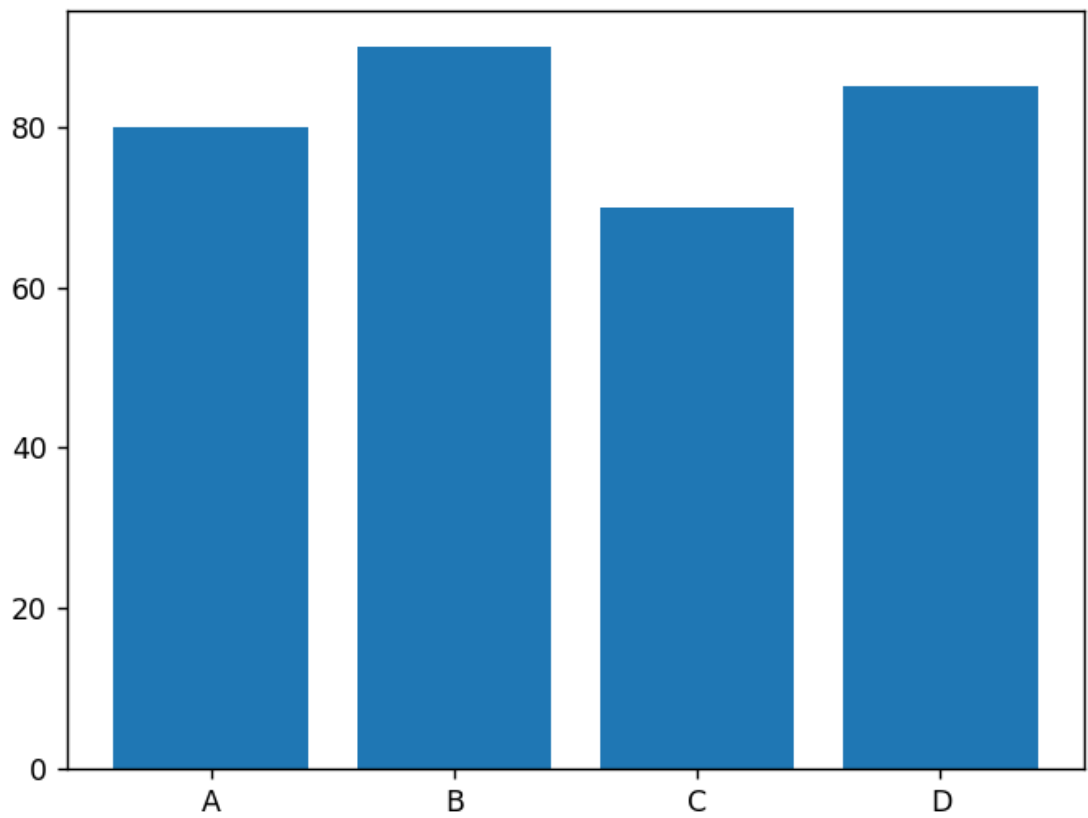
```

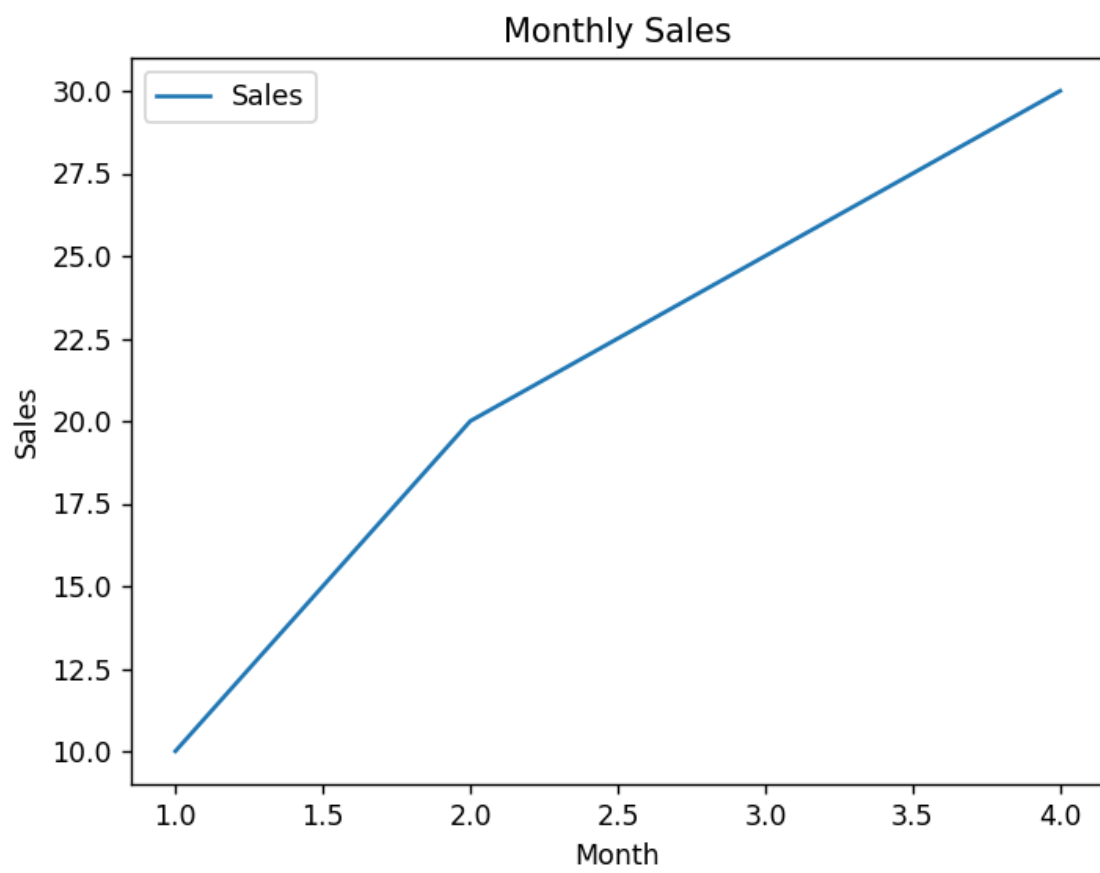
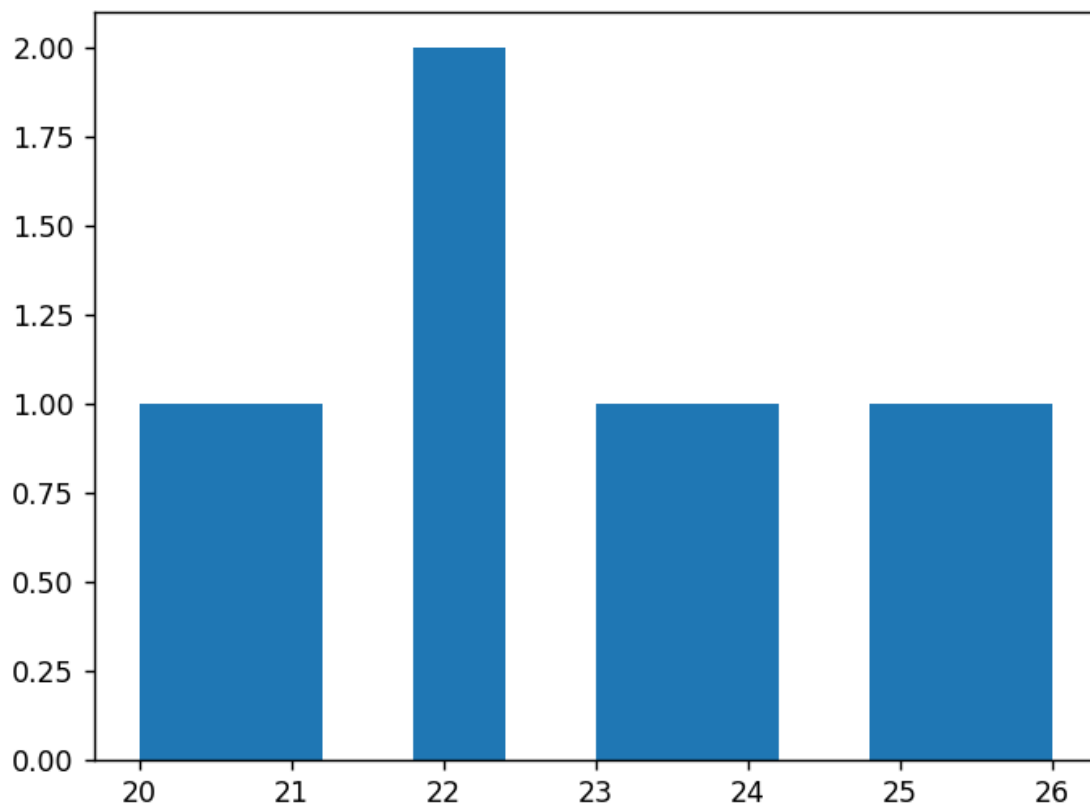
# Task 6
print("\nTask 6")
import matplotlib.pyplot as plt
x = [1,2,3,4]
y1 = [10,20,30,40]
y2 = [5,15,25,35]
plt.plot(x,y1,label="Product A")
plt.plot(x,y2,label="Product B")
plt.legend()
plt.show()

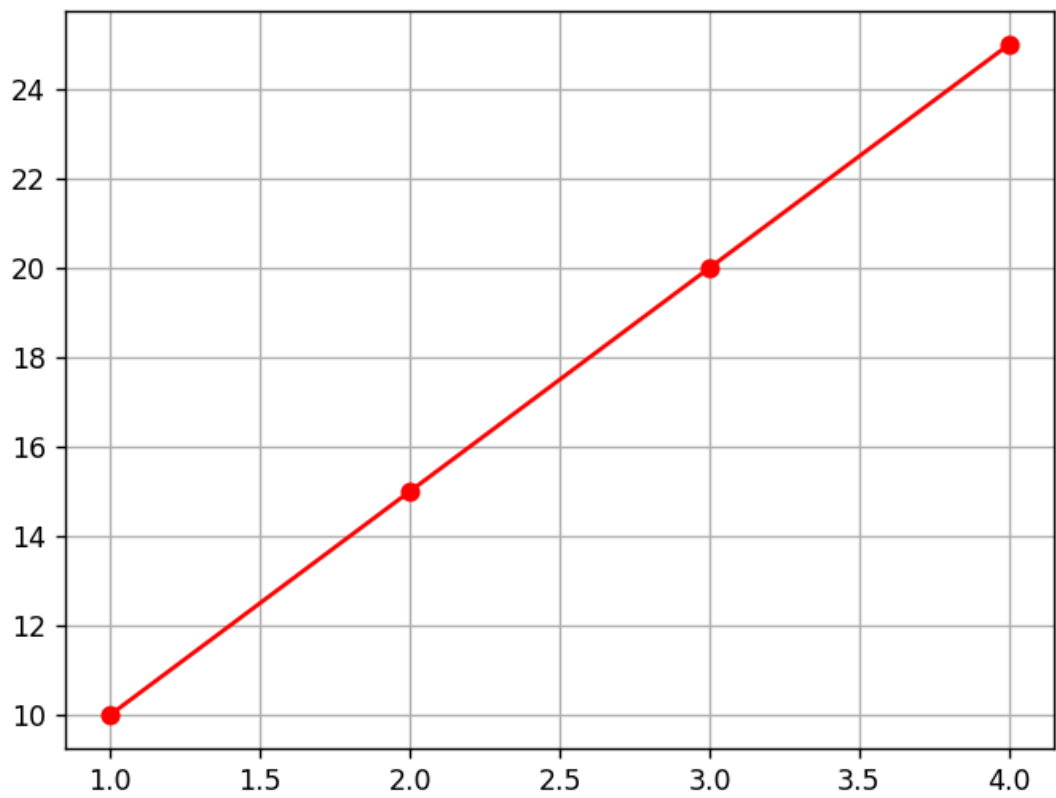
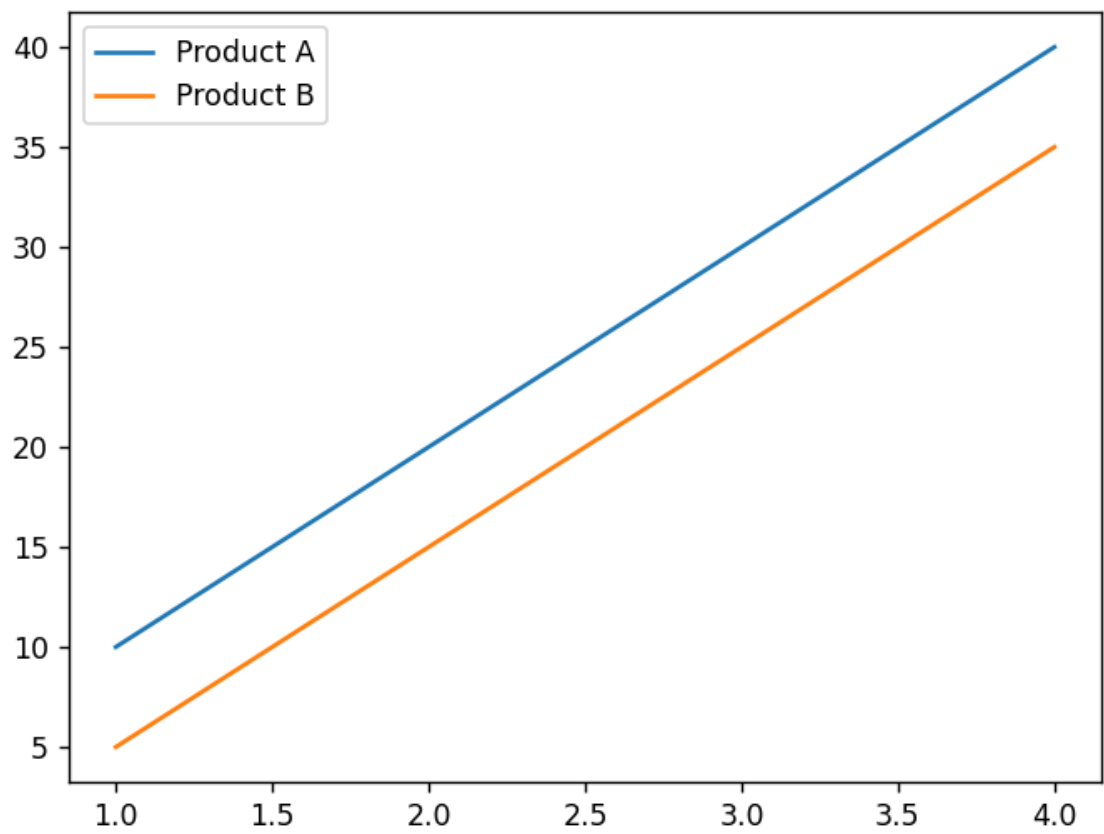
# Task 7
print("\nTask 7")
import matplotlib.pyplot as plt
x=[1,2,3,4]
y=[10,15,20,25]
plt.plot(x,y,color='red',marker='o')
plt.grid(True)
plt.show()

```









#Tkinter Tasks

Task 1

```
import tkinter as tk
window = tk.Tk()
window.title("Basic Window")
window.mainloop()
```

Task 2

```
import tkinter as tk
window = tk.Tk()
label = tk.Label(window, text="Hello")
label.pack()
button = tk.Button(window, text="Click")
button.pack()
window.mainloop()
```

Task 3

```
import tkinter as tk

def add():
    result.set(float(e1.get()) + float(e2.get()))

def sub():
    result.set(float(e1.get()) - float(e2.get()))

def mul():
    result.set(float(e1.get()) * float(e2.get()))

def div():
    result.set(float(e1.get()) / float(e2.get()))

window=tk.Tk()

e1=tk.Entry(window)
e2=tk.Entry(window)
e1.pack()
e2.pack()

result=tk.StringVar()

tk.Button(window, text="Add", command=add).pack()
tk.Button(window, text="Sub", command=sub).pack()
tk.Button(window, text="Mul", command=mul).pack()
tk.Button(window, text="Div", command=div).pack()

tk.Label(window, textvariable=result).pack()

window.mainloop()
```

Task 4

```
import tkinter as tk
window=tk.Tk()

tk.Label(window, text="Username").pack()
tk.Entry(window).pack()

tk.Label(window, text="Password").pack()
tk.Entry(window, show="*").pack()

tk.Button(window, text="Login").pack()

window.mainloop()
```

```
# Task 5
import tkinter as tk
from tkinter import messagebox

def show():
    messagebox.showinfo("Message", "Button Clicked")

window=tk.Tk()
tk.Button(window, text="Click", command=show).pack()
window.mainloop()

# Task 6
import tkinter as tk
window=tk.Tk()
textbox=tk.Entry(window)
textbox.pack()
window.mainloop()

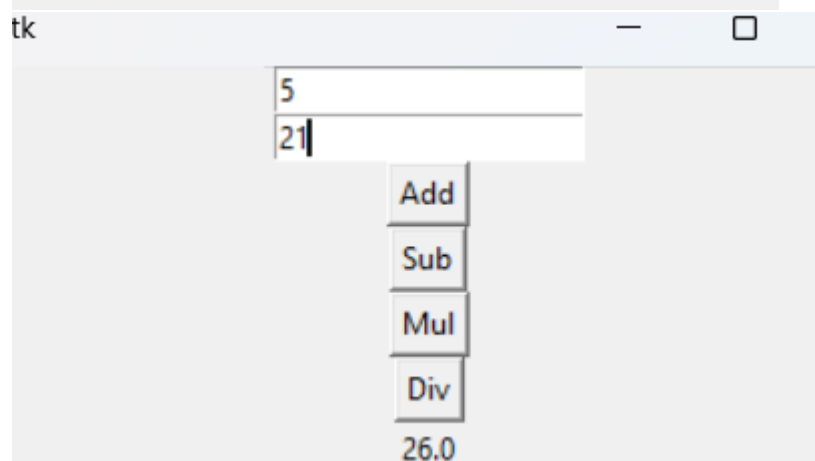
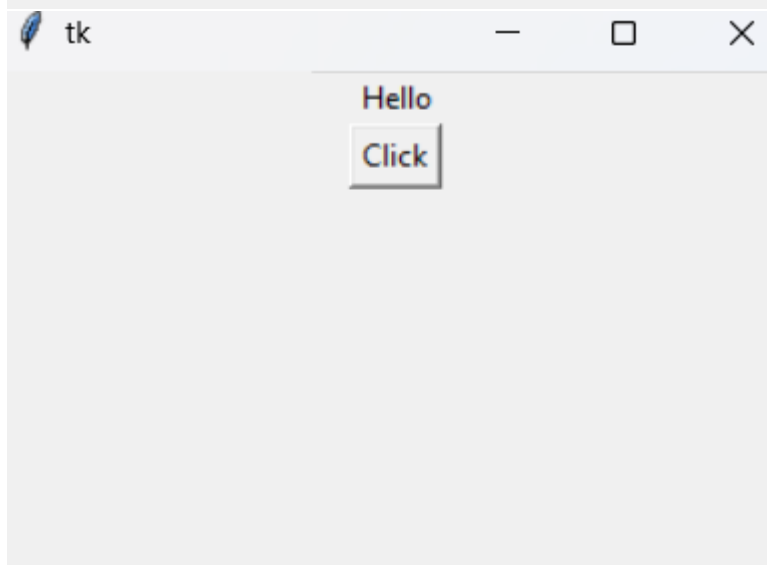
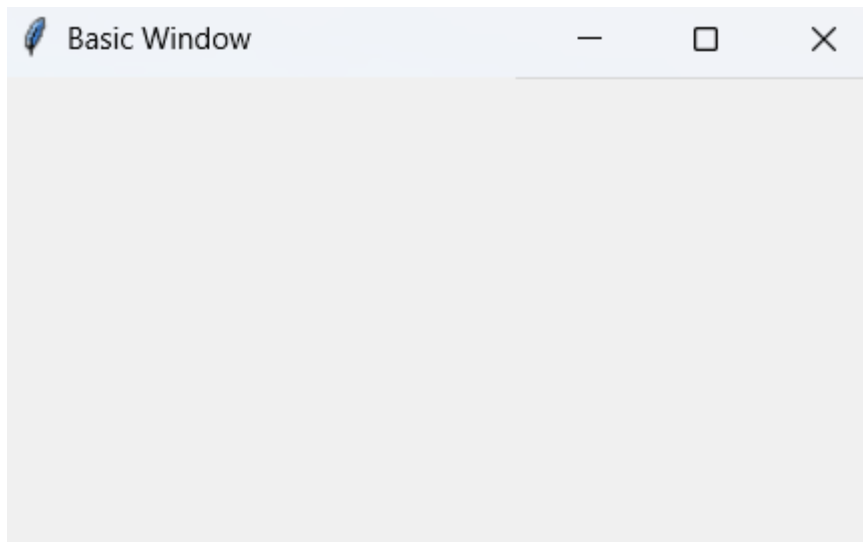
# Task 7
import tkinter as tk
window=tk.Tk()
text=tk.Text(window)
text.pack()
window.mainloop()

# Task 8
import tkinter as tk
window=tk.Tk()

menu=tk.Menu(window)
window.config(menu=menu)

file=tk.Menu(menu)
menu.add_cascade(label="File", menu=file)
file.add_command(label="New")

window.mainloop()
```



tk

Username

Subrahmanya

Password

Login

Click

Message

Button Clicked

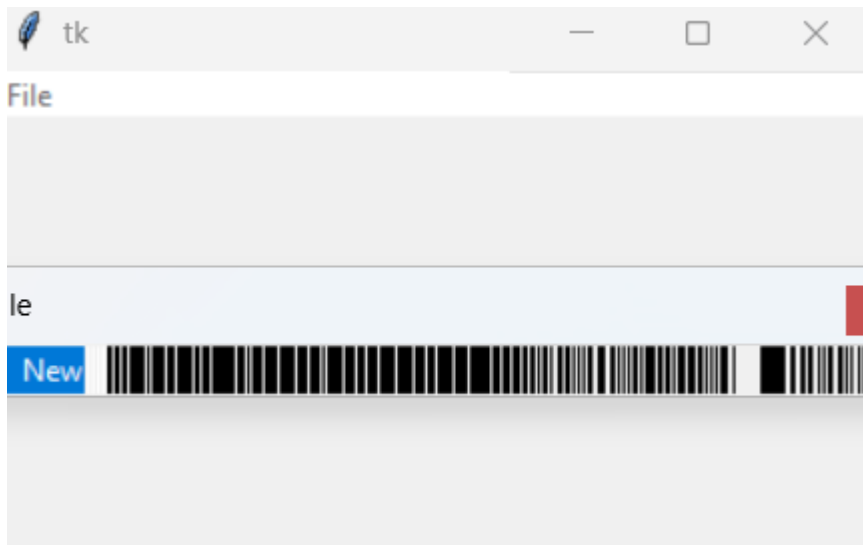
OK

tk

Good morning

tk

Hello Everyone, I'm Subrahmanya KR



```
#PyQt Tasks

import sys
from PyQt5.QtWidgets import QApplication, QWidget, QLabel, QPushButton, QLineEdit, QVBoxLayout

class MyApp(QWidget):

    def __init__(self):
        super().__init__()

        self.setWindowTitle("PyQt Tasks")
        self.setGeometry(200,200,300,200)

        layout = QVBoxLayout()

        self.label = QLabel("Login Form")

        self.username = QLineEdit()
        self.username.setPlaceholderText("Enter Username")

        self.password = QLineEdit()
        self.password.setPlaceholderText("Enter Password")
        self.password.setEchoMode(QLineEdit.Password)

        self.button = QPushButton("Login")

        self.button.clicked.connect(self.login)

        layout.addWidget(self.label)
        layout.addWidget(self.username)
        layout.addWidget(self.password)
        layout.addWidget(self.button)

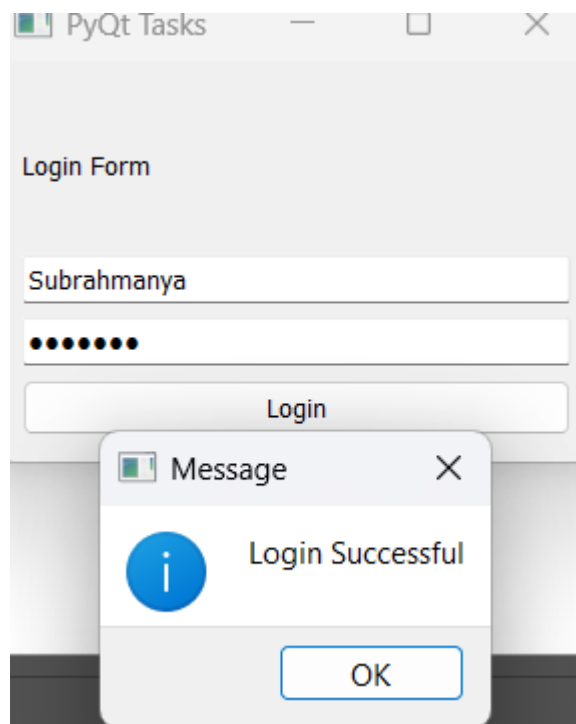
        self.setLayout(layout)

    def login(self):
        QMessageBox.information(self, "Message", "Login Successful")

app = QApplication(sys.argv)

window = MyApp()
window.show()

sys.exit(app.exec_())
```



```

#NLTK Tasks

# Task 1
print("Task 1")
import nltk
from nltk.tokenize import word_tokenize

text="Natural Language Processing is interesting"
print(word_tokenize(text))

# Task 2
print("\nTask 2")
import nltk
from nltk.tokenize import sent_tokenize

text="NLP is fun. It is used in AI."
print(sent_tokenize(text))

# Task 3
print("\nTask 3")
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

text="This is a simple sentence for stop word removal"
words=word_tokenize(text)

filtered=[w for w in words if w.lower() not in stopwords.words('english')]
print(filtered)

# Task 4
print("\nTask 4")
import nltk
from nltk.stem import PorterStemmer

ps=PorterStemmer()
words=["playing", "played", "plays"]
for w in words:
    print(ps.stem(w))
|

```

```

# Task 5
print("\nTask 5")

import nltk
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
print(lemmatizer.lemmatize("running", pos='v'))

# Task 6
print("\nTask 6")
import nltk
from nltk.tokenize import word_tokenize

text="The dog barked loudly"
words=word_tokenize(text)
print(nltk.pos_tag(words))

```

===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 33 TASK.1

```

Task 1
['Natural', 'Language', 'Processing', 'is', 'interesting']

Task 2
['NLP is fun.', 'It is used in AI.']

Task 3
['simple', 'sentence', 'stop', 'word', 'removal']

Task 4
play
play
play

Task 5
run

Task 6
[('The', 'DT'), ('dog', 'NN'), ('barked', 'VBD'), ('loudly', 'RB')]

```